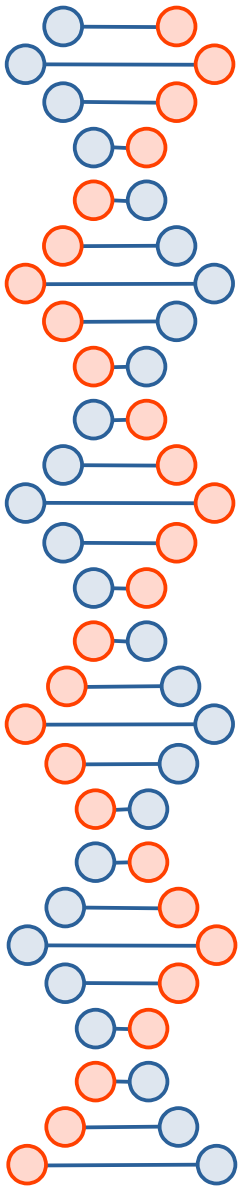


CAPÍTULO 2. Fundamentos de Machine Learning

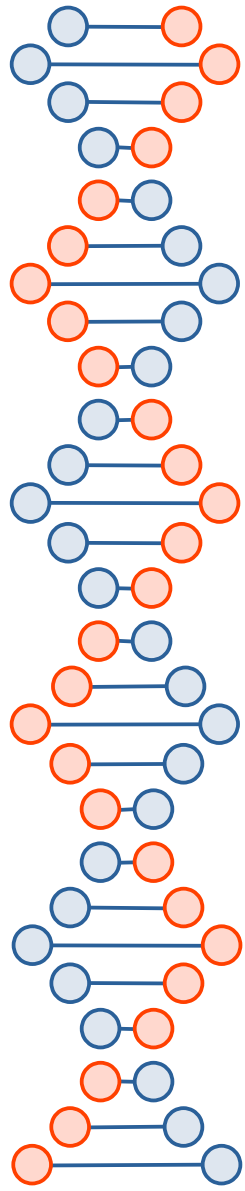
Parametric and Nonparametric Machine Learning Algorithms

Dr. Josafá Pontes
Machine Learning



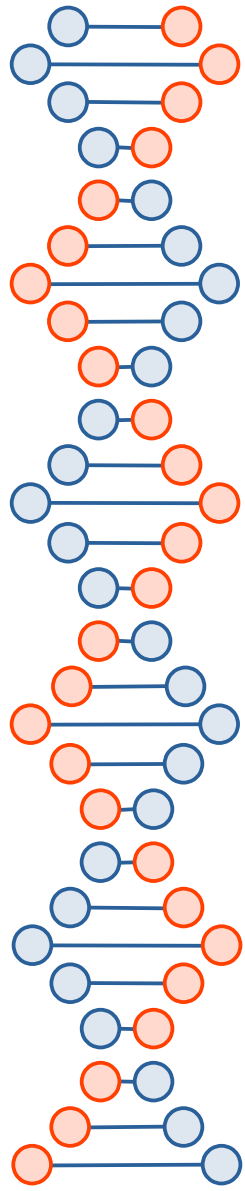
Outline

- Learn the difference between
 - a parametric machine learning algorithms
 - a nonparametric machine learning algorithm
- Classify among distinct algorithms



1. Parametric Machine Learning Algorithms (1)

- Assumptions can greatly simplify the learning process, but can also limit what can be learned.
- Algorithms that simplify the function to a known form are called parametric machine learning algorithms.

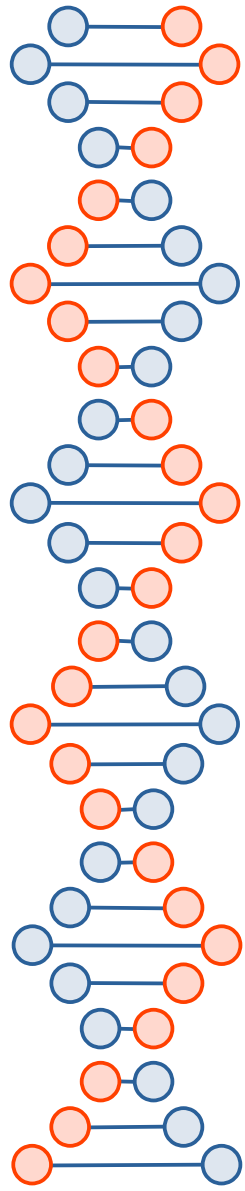


1. Parametric Machine Learning Algorithms (2)

A parametric model summarizes data with a set of parameters of fixed size.

No matter how much data you throw at a parametric model, it won't change its mind about how many parameters it needs.

— Artificial Intelligence: A Modern Approach, page 737

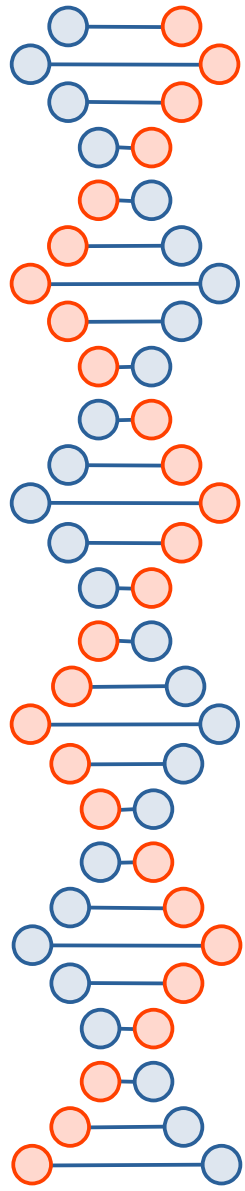


1. A parametric algorithm involve two steps:

- 1. Select a form for the function.
- 2. Learn the coefficients for the function from the training data.

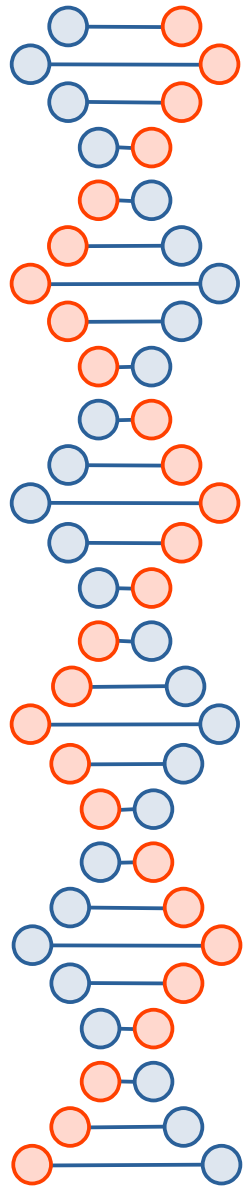
An easy to understand functional form for the mapping function is a line, as is used in linear regression:

$$B_0 + B_1 \times X_1 + B_2 \times X_2 = 0$$



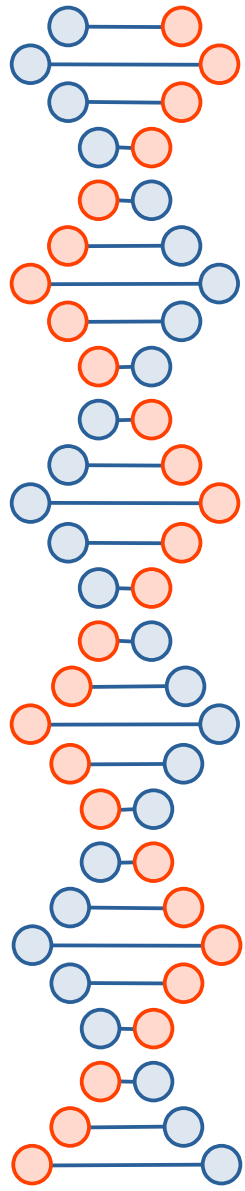
1. Examples of parametric machine learning algorithms

- Logistic Regression
- Linear Discriminant Analysis
- Perceptron



1. Benefits of Parametric Machine Learning Algorithms

- **Simpler:** These methods are easier to understand and interpret results.
- **Speed:** Parametric models are very fast to learn from data.
- **Less Data:** They do not require as much training data and can work well even if the fit to the data is not perfect.

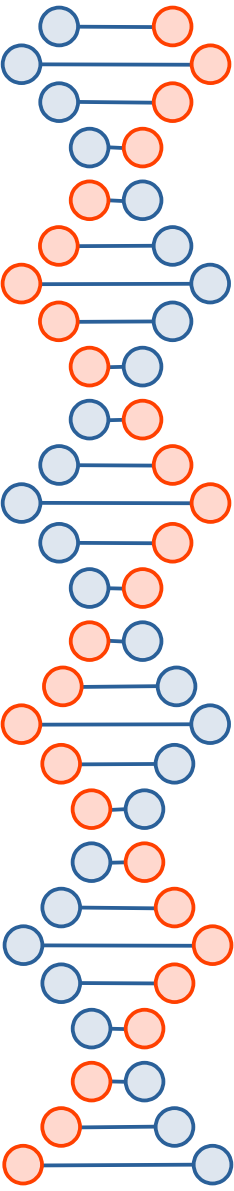


1. Limitations of Parametric Machine Learning Algorithms

- Constrained: By choosing a functional form these methods are highly constrained to the specified form.
- Limited Complexity: The methods are more suited to simpler problems.
- Poor Fit: In practice the methods are unlikely to match the underlying mapping function.

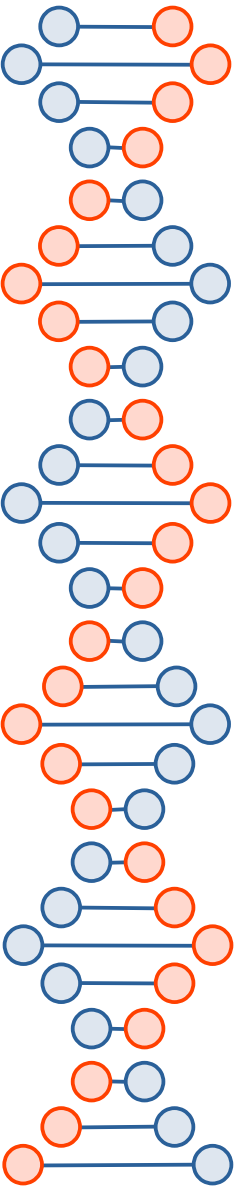
2. Nonparametric Machine Learning Algorithms (1)

- Nonparametric algorithms do not make strong assumptions about the form of the mapping function.
- By not making assumptions, they are free to learn any functional form from the training data.



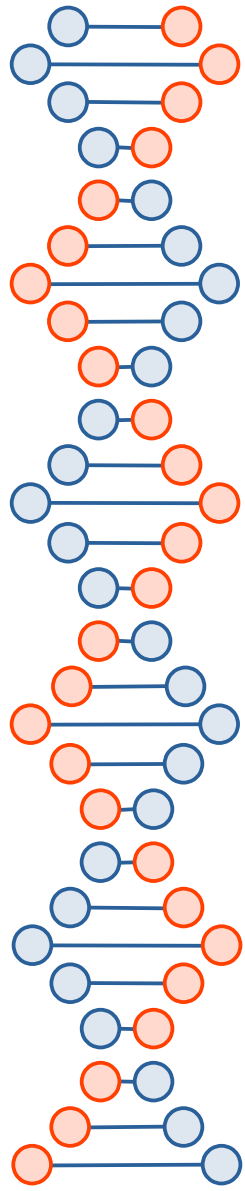
2. Nonparametric Machine Learning Algorithms (2)

- Nonparametric methods are good when you have a lot of data and no prior knowledge.
 - When you don't want to worry too much about choosing just the right features.
- Artificial Intelligence: A Modern Approach, page 757



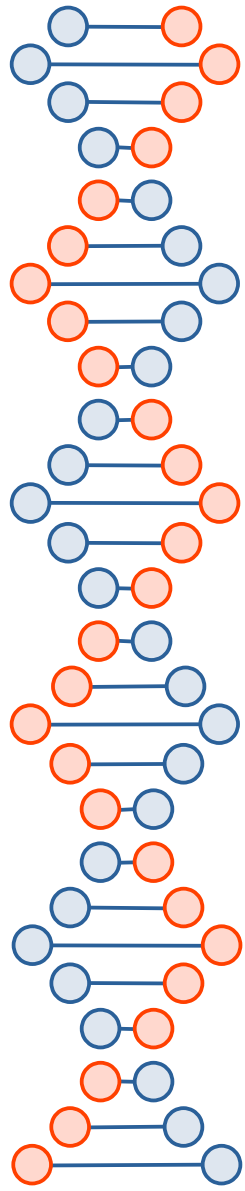
2. Nonparametric Machine Learning Algorithms (3)

- Nonparametric methods seek to best fit the training data in constructing the mapping function, whilst maintaining some ability to generalize to unseen data.
- They are able to fit a large number of functional forms.
- An easy to understand nonparametric model is the k-nearest neighbors (k-NN) algorithm.
- It makes predictions based on the k most similar training patterns for a new data instance.
- The method does not assume anything about the form of the mapping function, except that the patterns lead to similar outputs.



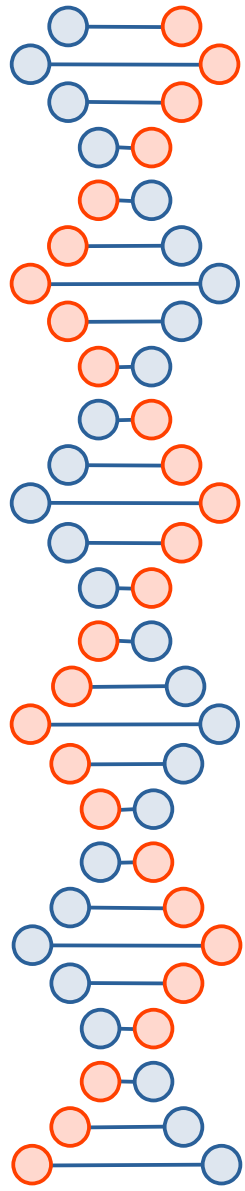
2. Examples of popular nonparametric machine learning algorithms

- kNN
- Decision Trees like CART and C4.5
- Naive Bayes
- Support Vector Machines
- Neural Networks



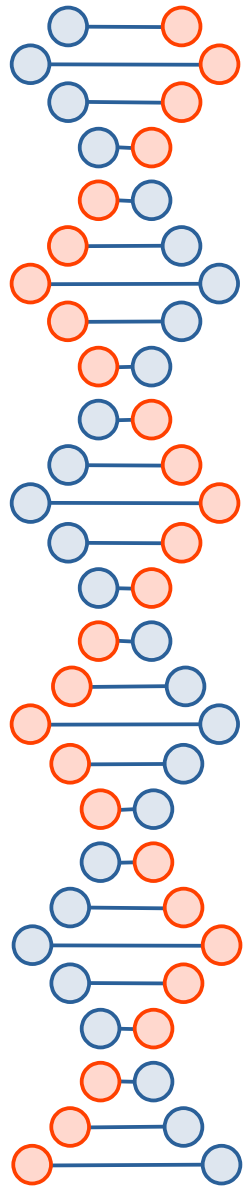
2. Benefits of Nonparametric Machine Learning Algorithms

- Flexibility: Capable of fitting a large number of functional forms.
- Power: No assumptions (or weak assumptions) about the underlying function.
- Performance: Can result in higher performance models for prediction.



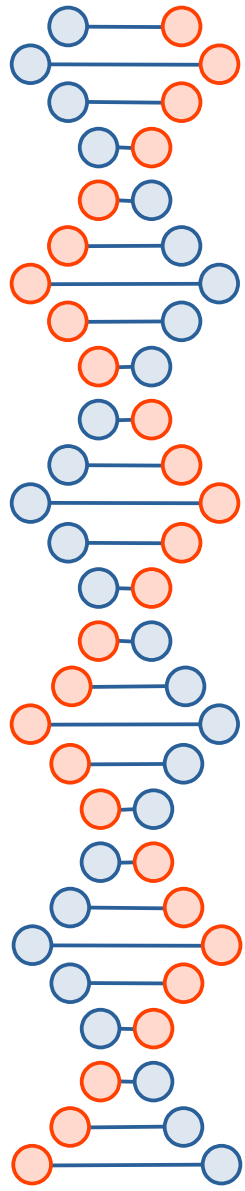
2. Limitations of Nonparametric Machine Learning Algorithms

- More data: Require a lot more training data to estimate the mapping function.
- Slower: A lot slower to train as they often have far more parameters to train.
- Overfitting: More of a risk to overfit the training data.
- It is harder to explain why specific predictions are made.

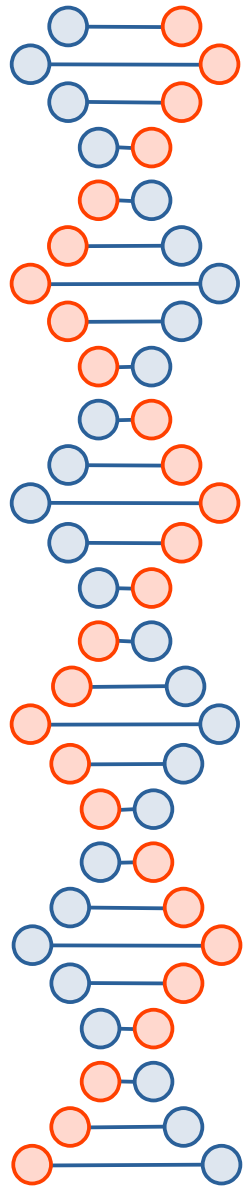


Comparison between Parametric vs Non-Parametric Machine Learning Algorithms

Feature	Parametric Algorithms	Non-Parametric Algorithms
Definition	Have a fixed number of parameters.	The number of parameters grows with the training data.
Examples	Linear Regression, Logistic Regression, LSTM, CNN, MLP	k-NN, Decision Trees, Random Forests, SVM (with some kernels)
Model Complexity	Fixed after training (defined by architecture).	Increases with more data.
Training Speed	Typically faster to train.	Slower due to complexity increasing with data.
Prediction Speed	Typically faster (just apply learned weights).	Slower, especially for instance-based models like k-NN.
Flexibility	Less flexible; may underfit if the model is too simple.	More flexible; better at capturing complex patterns.
Data Requirements	Require fewer data to generalize well (if well-regularized).	Often require more data to perform reliably.
Interpretability	Often more interpretable (especially linear models).	Often harder to interpret (e.g., large trees or kernel SVMs).

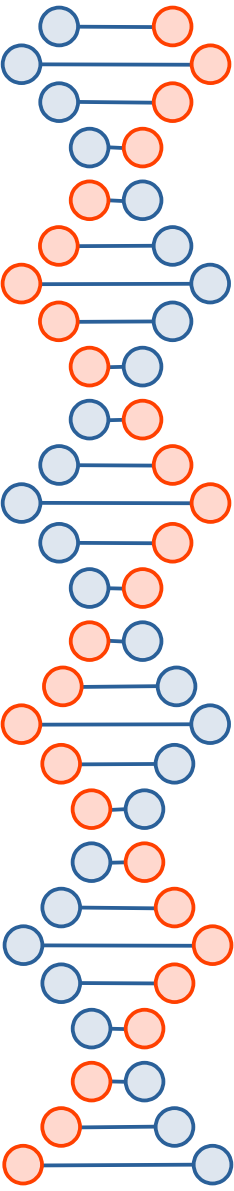


Are Deep Learning models such as LSTMs or
Convolutional Neural Networks Parametric
Machine Learning Algorithms?



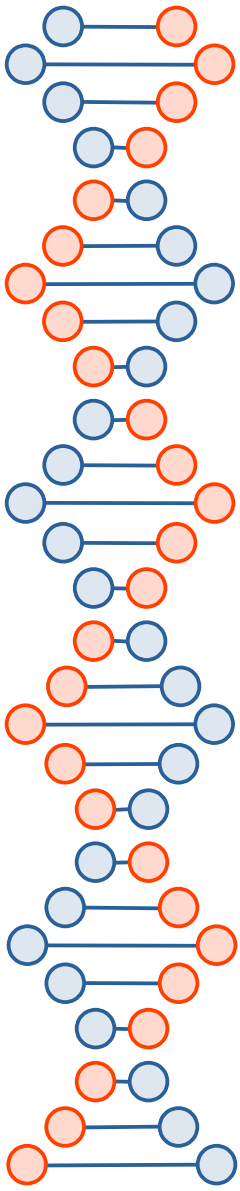
Considerations

- Parametric algorithms are defined as models that:
- Have a fixed number of parameters (though possibly very large).
- Learn these parameters from training data.
- Once trained, do not grow in complexity regardless of the amount of data.



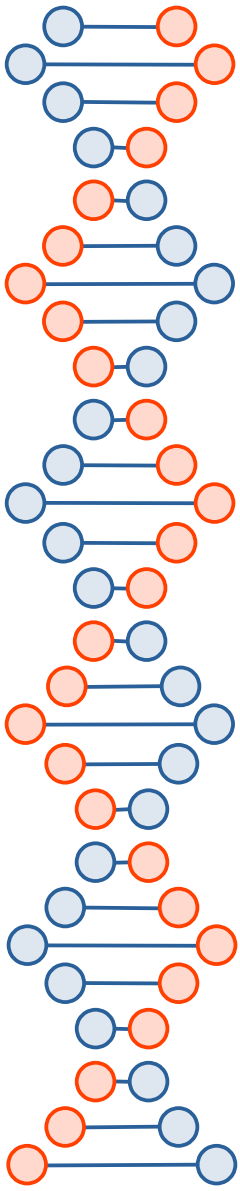
More considerations

- LSTMs (Long Short-Term Memory networks) have a predefined architecture with weights (parameters) in gates (input, forget, output) and cell states. These weights are learned during training, and their number doesn't change with more data.
- CNNs (Convolutional Neural Networks) also have a fixed number of filters (kernels) and weights in convolutional and fully connected layers.
- Even though these models can be very large and complex, the number of parameters is fixed once the architecture is defined — so they are still parametric by definition.



Answer

- Deep learning models (like CNNs and LSTMs) fall under parametric models because their architecture is defined ahead of time and does not change based on data size.
- Non-parametric models adapt their structure or parameters as more data becomes available.



Summary

- You have discovered the difference between parametric and nonparametric machine learning algorithms.
- Parametric methods make large assumptions about the mapping of the input variables to the output variable.
- They are faster to train and require less data but may not be as powerful.
- Nonparametric methods make few or no assumptions about the target function.
- They require a lot more data, are slower to train and have a higher model complexity but can result in more powerful models.